

Deep learning projects

May 8, 2026

In the underlying problems, the cantilever beam structure is made of centered rectangular unit cells. Each strut has different thickness values. The overall task of the project is to understand the effect of the thickness distribution on the von Mises stress.

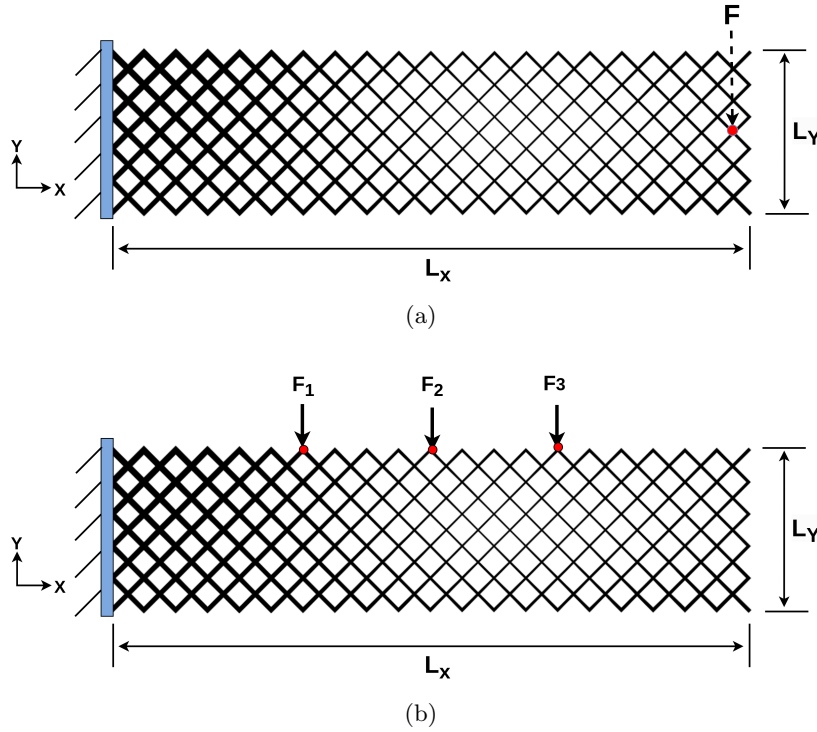


Figure 1: Description of the underlying cantilever beam problem of the centered rectangular unit cells, having (a) a point load at the mid of the right edge of the beam, and (b) 3 point loads at the top edge of the beam.

The thickness values are randomly generated at the nodes of the structure, the nodes for your reference are shown in Fig. 2. The thickness of each strut is

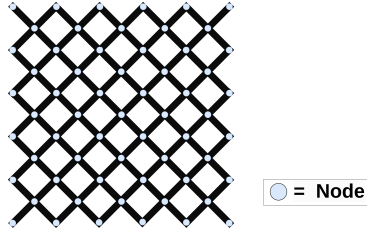


Figure 2: FGL domain is discretized into the nodes, and each node is assigned a geometric parameter value.

taken as a linear variation based on the nodal values. For problem 1a, we have generated numerous random strut-thickness distribution profiles and obtained the Von Mises stress field for each strut-thickness distribution. The load location and magnitude have not been varied, the only variation across the dataset is the thickness distribution. Data file of the problem 1 includes:

- An "output.xlsx" file containing 5000 rows and 226 columns. Each row stores data for one strut-thickness profile.
- A "stress" folder containing 5000 text files. Each file corresponds to one row of the "output.xlsx" file. In each text file:
 - Column 1 represents the Von Mises stress values,
 - Column 2 represents the deformed X-coordinates,
 - Column 3 represents the deformed Y-coordinates.
- A "cord.txt" file containing the X- and Y-coordinates of the undeformed structure.

For problem 1b, we have varied point-load locations (F_1 , F_2 , and F_3) as well as magnitude along with the strut-thickness distributions. For each case, the corresponding Von Mises stress field was obtained. The dataset contains the same three files as in Problem-1. In addition, an extra folder named "loaddata" is provided. The "loaddata" folder contains 5000 text files, each containing information about the applied loads and the corresponding nodes. In each text file of the loaddata folder:

- Column 1 represents the node number where the load is applied.
- Column 3 represents the magnitude of the load applied at the corresponding node.

Q1.

- Predict the maximum Von Mises stress for the given material distribution for Problem 1a.
- Calculate the derivatives of the maximum Von Mises stress with respect to volume fraction. You should use the backpropagation (with the help of the Keras library) for Problem 1a.
- Run an optimization algorithm to find the material distribution that provides the minimum value of the maximum Von Mises stress. You can use the simplest optimization approach.

Q2: Problem 1a.

- Predict the Von Mises stress field for a given volume fraction distribution using UNet.
- Now train another Unet, that provides volume fraction distribution for a given Von Mises stress distribution.
- Compare the complexity of the above two problems, for example in terms of the data requirements, stability.

Q3.

- Predict the maximum Von Mises stress for the given material distribution for a fixed boundary condition problem 1a.
- Now, predict the maximum Von Mises stress for a given material distribution, and load distribution problem 1b.
- Compare the complexity of the both the approaches.

Q4: problem 1a.

- Approach-1: Predict the maximum Von Mises stress for the given volume fraction distribution.
- Approach-2: Now, carry out the dimensionality reduction of the input data (volume fraction distribution). Now, carry out the prediction of the maximum Von Mises stress, by taking the reduced dimensional coordinates as input.
- Compare the data requirement of the approach-1 vs approach-2.

Q6 problem 1a.

- Approach-1: Predict the maximum Von Mises stress for the given volume fraction distribution using CNN.

- Approach-2: Predict the Von Mises stress field for a given volume fraction distribution using UNet. Use this approach to finally predict the maximum Von Mises stress.
- Which approach is superior, compare the pros and cons. For example, compare the data requirement of the approach-1 vs approach-2.

Q 7 : problem 1a.

- Approach-1: Predict the maximum Von Mises stress for the given volume fraction distribution using CNN.
- Approach-2: Train a CNN, to predict whether the maximum Von Mises stress is in 0-10 percentile, 11-25 percentile, 26-50 percentile, 51-100 percentile. Here, the 0-10 percentile designs are the ones having the least maximum Von Mises stress.
- Which approach is superior, compare the pros and cons. For example, compare the data requirement of the approach-1 vs approach-2.

Q8 problem 1a.

- Predict the maximum Von Mises stress for the given material distribution for Problem 1a.
- Predict the stress distribution for a given material distribution. That means the spatial coordinate will also act as input.
- Compare the complexity of the above two problems in terms of network complexity and data requirement.

Q 9 : problem 1a.

- Approach-1: Predict the maximum Von Mises stress for the given volume fraction distribution using any network.
- Approach-2: Train a NN, to predict whether the maximum Von Mises stress is in 0-10 percentile, 11-25 percentile, 26-50 percentile, 51-100 percentile. Here, the 0-10 percentile designs are the ones having the least maximum Von Mises stress.
- Which approach is superior, compare the pros and cons. For example, compare the data requirement of the approach-1 vs approach-2.